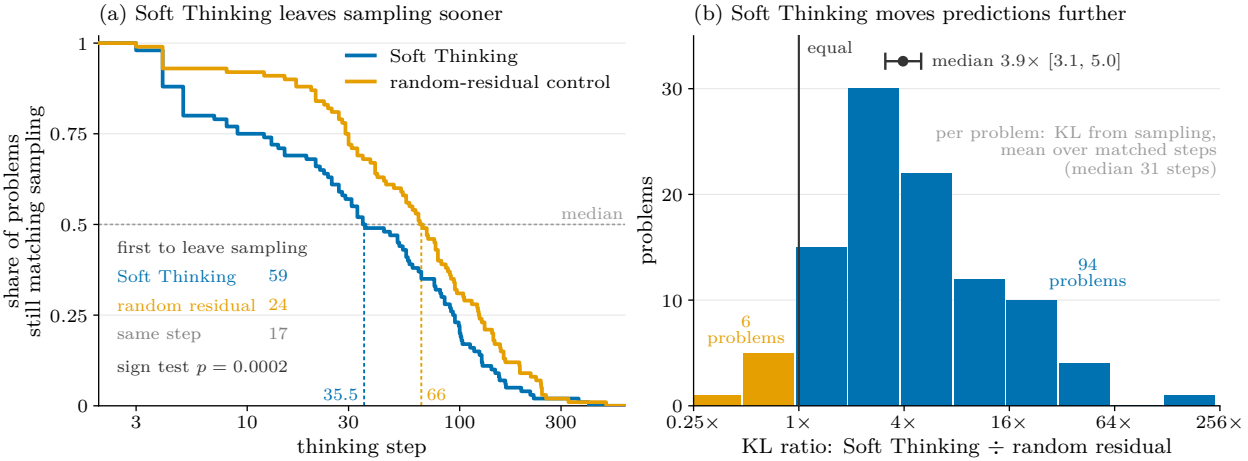

(a) Soft Thinking leaves sampling sooner
Soft Thinking
random-residual control
share of problems still matching sampling
1
0.75
0.5
0.25
0
median
first to leave sampling
Soft Thinking 59
random residual 24
same step 17
sign test $p = 0.0002$
35.5
66
3
10
30
100
300
thinking step
(b) Soft Thinking moves predictions further
equal
median 3.9× [3.1, 5.0]
per problem: KL from sampling, mean over matched steps (median 31 steps)
problems
30
20
10
0
6 problems
94 problems
0.25×
1×
4×
16×
64×
256×
KL ratio: Soft Thinking ÷ random residual